Logarithms

1

a $\log_5 25 = 2$

b $\log_3 81 = x$

c $\log_3 3 = 1$

d $\log_2 1 = 0$

e $\log_{4.4} 1 = 0$

f $\log_{25} 125 = 1.5$

g $\log_4 32 = \frac{5}{2}$

h $\log_4 0.0625 = -2$

i $\log_4 \frac{1}{64} = -3$

j $\log_x 64 = 1.5$

2

a $\log_{10} 100 = 2$

b $\log_{10} \left(\frac{1}{100}\right) = -2$

c $\log_{10} \left(\sqrt{10}\right) = \frac{1}{2}$

d $\log_{10} \left(\frac{1}{\sqrt{10}}\right) = -\frac{1}{2}$

3

a $4^2 = 16$

b $5^1 = 5$

c $8^0 = 1$

d $2^{-3} = 0.125$

e $3^{-1} = \frac{1}{3}$

f $(5.8)^2 = 33.64$

g $\left(\frac{1}{3}\right)^{-2} = 9$

h $x^5 = 32$

i $8^6 = x$

4

a $10^4 = 10\,000$

b $10^{\frac{1}{2}} = \sqrt{10}$

c $10^{-4} = \frac{1}{10\,000}$

d $10^{-\frac{1}{2}} = \frac{1}{\sqrt{10}}$

5 True

6 1

7

a 4

b -2

c -1

d 0

e $\frac{1}{2}$

f -2

g -1

h $\frac{1}{3}$

i 3

j $\frac{1}{4}$

k 1

l $\frac{1}{8}$

m $-\frac{4}{3}$

n $\frac{15}{4}$

8 0.713

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9

a 1 **b** 2

c Between 1 and 2

10

a Between 3 and 4 **b** 3.9490

11

a No **b** No **c** Yes, 0.30 **d** Yes, 1.26

12 3

13

a 6 **b** 0 **c** 8 **d** -3

14 True

15 $x = 25$

16 1.32

17 $k = 4$

18

a 4 **b** $x = 1$

19

a

i $x = \log_{10} 790$ **ii** 2.90

b

i $x = \log_{10} 0.0003$ **ii** -3.52

20

a -1 **b** 9 **c** -2 **d** $\frac{1}{2}$

e $-\frac{1}{2}$

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